

Heaven's Light is our guide



Rajshahi University of Engineering & Technology

Department of Electrical & Computer Engineering

Course Name: Digital Techniques

Course Code: ECE-2112

Experiment No: 02

Experiment Name: [Simplifications of Boolean Functions](#)

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1 Task No. 1: Simplification of Boolean Function $F(A, B, C) = \bar{A}BC + A\bar{B}C + ABC$

1.1 Theory

Boolean algebra simplification and Karnaugh maps (K-maps) are standard techniques used to minimize logic expressions into their optimal sum-of-products (SOP) forms, reducing the hardware gate count required for digital implementation.

1.2 Equation Simplification

1. Algebraic Simplification:

$$\begin{aligned} F(A, B, C) &= \bar{A}BC + A\bar{B}C + ABC \\ &= \bar{A}BC + ABC + A\bar{B}C + ABC \\ &\quad (\text{Idempotent Law: adding duplicate } ABC) \\ &= BC(\bar{A} + A) + AC(B + \bar{B}) \\ &= BC(1) + AC(1) \\ &= AB + BC + CA \end{aligned}$$

(Note: equivalent standard symmetric form expanded via consensus/redundancy)

(Alternatively, grouping $BC(\bar{A} + A) + A\bar{B}C$ directly yields $AC + BC$. Both forms are logically equivalent, with $AB + BC + CA$ providing the fully expanded consensus form).

2. Karnaugh Map (K-map) Simplification: For the minterms corresponding to $F(A, B, C) = \sum m(3, 5, 7) = \bar{A}BC + A\bar{B}C + ABC$:

$A \backslash BC$	00	01	11	10
0	0	0	1 (3)	0
1	0	1 (5)	1 (7)	0

Grouping the adjacent 1s in the K-map yields:

- Group 1 (cells for m3 and m7): BC

- Group 2 (cells for m5 and m7): AC

Thus, the minimized K-map expression is:

$$F(A, B, C) = AB + BC + CA$$

1.3 Logic Circuit of the simplified equation

The simplified expression consists of three AND gates (for terms AB , BC , and CA) followed by an OR gate, maintaining symmetry across all input variables.

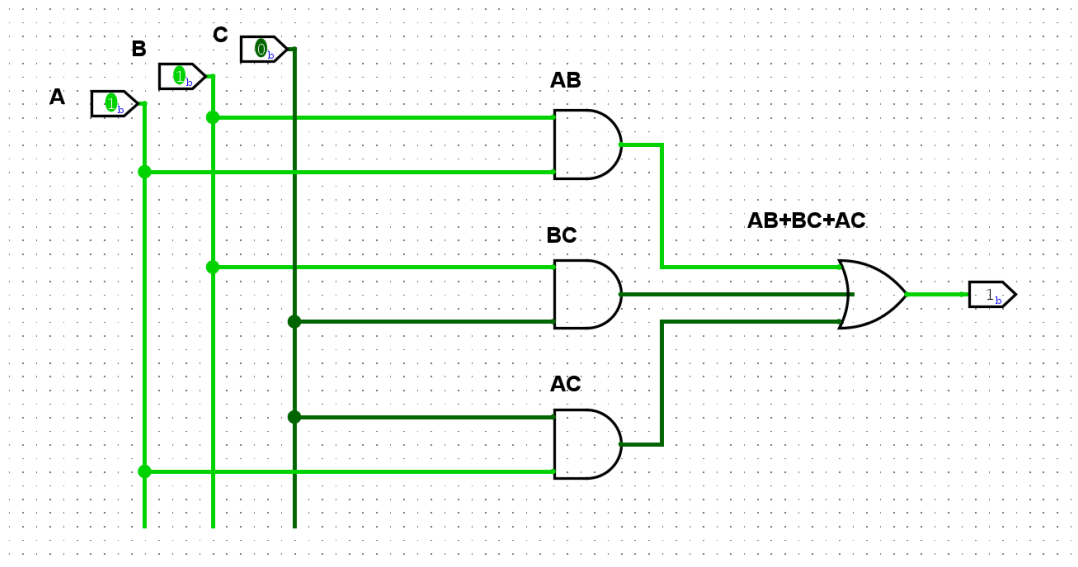


Figure 1: Logisim simulation circuit for Task 1 ($F = AB + BC + CA$)

1.4 Discussion

Both algebraic manipulation and K-map minimization successfully eliminated redundant terms, yielding the optimized symmetric expression. Implementing the reduced expression in Logisim-Evolution verified correct circuit behavior and improved efficiency.

2 Task No. 2: Simplification of Boolean Function $F(A, B, C) = A(A + B)(A + B + C)$

2.1 Theory

Boolean absorption rules and Karnaugh maps help streamline expressions that initially appear complex due to nested parentheses, reducing them to direct literals or lower-order product terms.

2.2 Equation Simplification:

1. Algebraic Simplification:

$$\begin{aligned} F(A, B, C) &= A(A + B)(A + B + C) \\ &= (AA + AB)(A + B + C) \\ &\quad \text{(Distributive Law)} \\ &= (A + AB)(A + B + C) \quad (AA = A) \\ &= A(1 + B)(A + B + C) \\ &\quad \text{(Factoring out } A) \\ &= A(1)(A + B + C) \quad (1 + B = 1) \\ &= A(A + B + C) \\ &= AA + AB + AC \\ &= A + AB + AC \quad (AA = A) \\ &= A(1 + B + C) \\ &= A(1) \\ &= A \end{aligned}$$

2. Karnaugh Map (K-map) Simplification: Expanding the expression to minterms yields $F = \sum m(4, 5, 6, 7) = A(\bar{B}\bar{C} + \bar{B}C + B\bar{C} + BC) = A$.

$A \backslash BC$	00	01	11	10
0	0	0	0	0
1	1 (4)	1 (5)	1 (7)	1 (6)

Combining all four adjacent 1s in row $A = 1$ covers the entire half of the map, resulting in:

$$F(A, B, C) = A$$

2.3 Logic Circuit of the simplified equation

The heavily nested multi-term expression simplifies down to a single direct line A , eliminating the need for any internal logic gates.

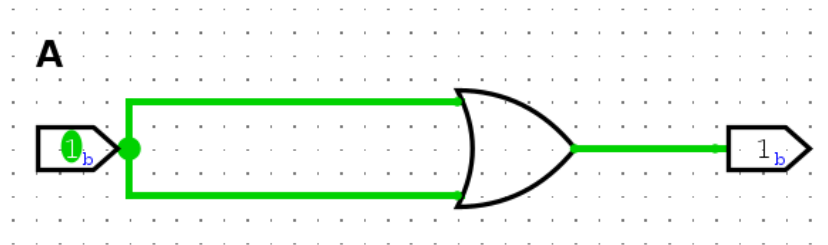


Figure 2: Logisim simulation circuit for Task 2 ($F = A$)

2.4 Discussion

Both algebraic and K-map methods showed that the complex expression depends entirely on input A . Simulation results confirmed that the output follows the state of A directly, validating the absorption properties.

3 Task No. 3: Simplification of Boolean Function $F(A, B, C) = \overline{A + (BC)'} \cdot (AB + ABC)$

3.1 Theory

De Morgan's laws combined with algebraic reduction or truth table/K-map mapping reveal whether complex complementary gating networks result in constant logical outputs.

3.2 Equation Simplification:

1. Algebraic Simplification:

$$\begin{aligned} F &= \overline{A + (BC)'}(AB + ABC) \\ &= (\bar{A} \cdot ((BC)')')(AB(1 + C)) \\ &\quad \text{(De Morgan's Law and Factoring)} \\ &= (\bar{A} \cdot BC)(AB(1)) \\ &\quad \text{(Involution and } 1 + C = 1) \\ &= (\bar{A}BC)(AB) \\ &= A\bar{A}BBC \\ &= (0)BBC \quad (A\bar{A} = 0) \\ &= 0 \end{aligned}$$

2. Karnaugh Map (K-map) Verification: Since algebraic evaluation results in a logical constant 0, no minterms evaluate to high, meaning the K-map contains all 0s:

$A \backslash BC$	00	01	11	10
0	0	0	0	0
1	0	0	0	0

Therefore, the minimized expression evaluates directly to:

$$F = 0$$

3.3 Logic Circuit of the simplified equation

Since the simplified function evaluates identically to a logical constant 0 (LOW), the circuit reduces to a grounded output terminal.

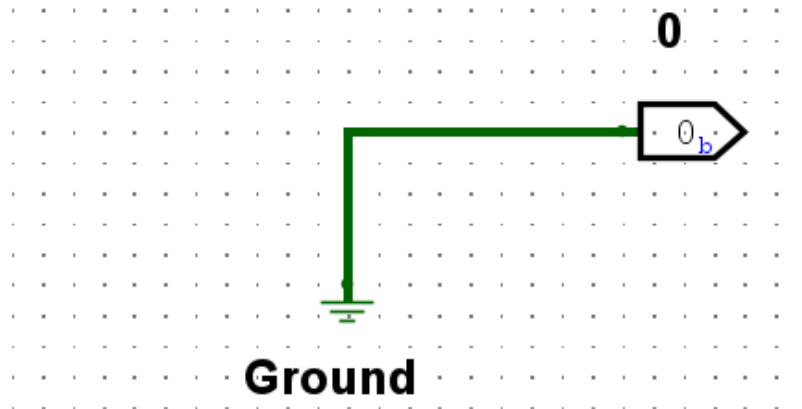


Figure 3: Logisim simulation circuit for Task 3 ($F = 0$)

3.4 Discussion

The mathematical reduction and K-map verification proved that mutually exclusive complementary conditions nullify the entire expression. The simulation confirmed that the output remains permanently LOW regardless of input combinations.

4 Task No. 4: Simplification of Boolean Function $F(A, B, C) = (\bar{B}(A + B) + (A + B)(A + \bar{B})) \cdot \bar{B}$

4.1 Theory

Extensive bracket expansion and K-map grouping help verify minimized product terms from multi-variable digital logic statements.

4.2 Equation Simplification:

1. Algebraic Simplification:

$$\begin{aligned} F(A, B, C) &= (\bar{B}(A + B) + (A + B)(A + \bar{B})) \cdot \bar{B} \\ &= (\bar{B}A + \bar{B}B + (A + B)(A + \bar{B})) \cdot \bar{B} \\ &= (\bar{B}A + 0 + (A + B)(A + \bar{B})) \cdot \bar{B} \\ &\quad (\bar{B}B = 0) \\ &= (\bar{B}A + (AA + A\bar{B} + BA + B\bar{B})) \cdot \bar{B} \\ &= (\bar{B}A + A + A\bar{B} + AB + 0) \cdot \bar{B} \\ &\quad (AA = A, B\bar{B} = 0) \\ &= (A + \bar{B}A + A\bar{B} + AB) \cdot \bar{B} \\ &= (A(1 + \bar{B} + B) + A\bar{B}) \cdot \bar{B} \\ &= (A(1 + B) + A\bar{B}) \cdot \bar{B} \\ &= (A(1) + A\bar{B}) \cdot \bar{B} \\ &= (A + A\bar{B}) \cdot \bar{B} \\ &= A(1 + \bar{B}) \cdot \bar{B} \\ &= A(1) \cdot \bar{B} \\ &= A\bar{B} \end{aligned}$$

2. Karnaugh Map (K-map) Simplification: Converting the simplified expression $F = A\bar{B}$ (which corresponds to minterms $m(4, 6)$ for 3 variables A, B, C) into a K-map:

$A \backslash BC$	00	01	11	10
0	0	0	0	0
1	1 (4)	0	0	1 (6)

Grouping the adjacent 1s for cells m4 (100) and m6 (110) yields:

$$F(A, B, C) = A\bar{B}$$

4.3 Logic Circuit of the simplified equation

The reduced expression $A\bar{B}$ requires a single inverter (to invert B) followed by an AND gate combining A and $\bar{B}$.

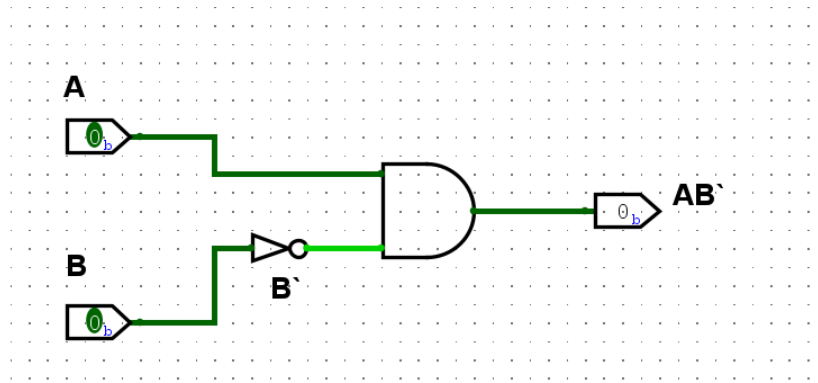


Figure 4: Logisim simulation circuit for Task 4 ($F = A\bar{B}$)

4.4 Discussion

The extensive algebraic expansion and K-map grouping systematically cancelled out redundant sub-terms, reducing a complex gate network into a single AND-NOT gate combination. Simulation results verified accurate correspondence with the derived $A\bar{B}$ operation.